

Web Technology



Introduction to Server-Side Programming

Server-Side Programming

1. **Tools and Programming Languages to**
 - **Design**
 - **Build**
 - **Maintain**
2. **Also Called Backend Development**

Server-Side Tasks

- **Process User Input and Request**
- **Querying and Managing Databases**
- **Managing States and Sessions**
- **Serving Dynamic Content and Responses**
- **Handling Authorization and Authentication**
- **Interact with Other Servers**
- **Read/ Write/ Access a File on Server**
- **Sending Notification and Communications**
- **Error Handling and Logging**
- **Security and Maintenance**

Server-Side Tasks

- **Process User Input and Request – Login Credentials, Formatting**
- **Querying and Managing Databases – CRUD Operations**
- **Managing States and Sessions – Store Information – Login Information**
- **Serving Dynamic Content and Responses – Send Personalized Data**
- **Handling Authorization and Authentication – Access Control/ Validate Login**
- **Interact with Other Servers – Fetching Data External API**
- **Read/ Write/ Access a File on Server – Generating Reports**
- **Sending Notification and Communications – SMS, Email, Push Notification**
- **Error Handling and Logging – Detect, Manage, and Log Errors**
- **Security and Maintenance - Encryption**

Types of Servers - Functionality

1. **Static**
2. **Dynamic**

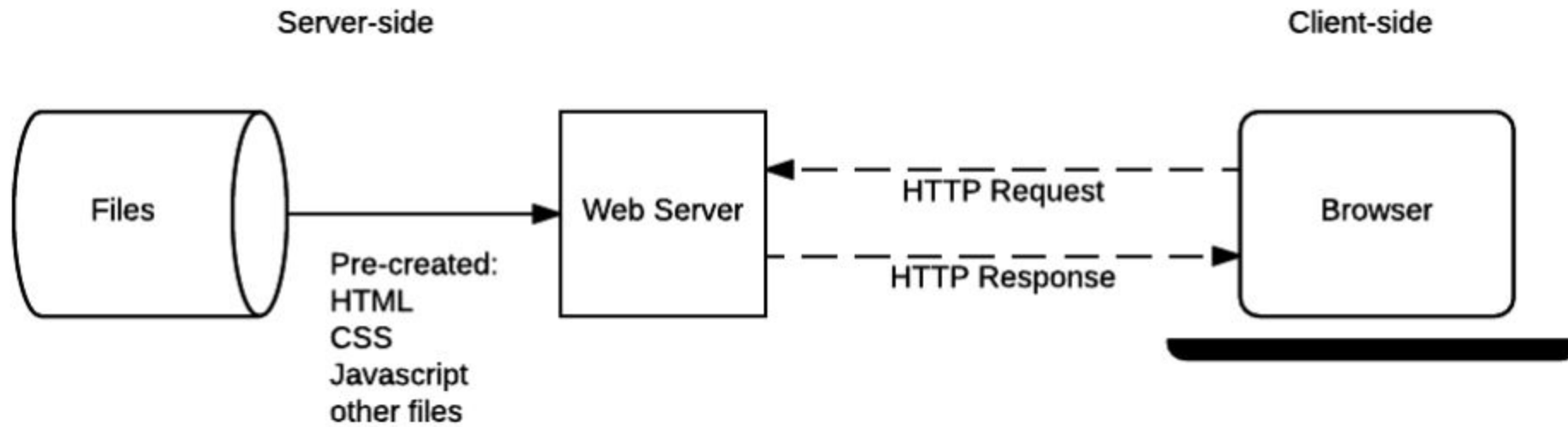
Types of Servers - Functionality

1. Static

- **HTTP Request and Response**
- **Gets Hard-Coded Content Stored from Server**
- **Suitable When Content Doesn't Change Frequently**
- **Usually No Heavy Backend Databases**
- **Eg: Blogs, Documentation, E-Commerce Platforms**

2. Dynamic

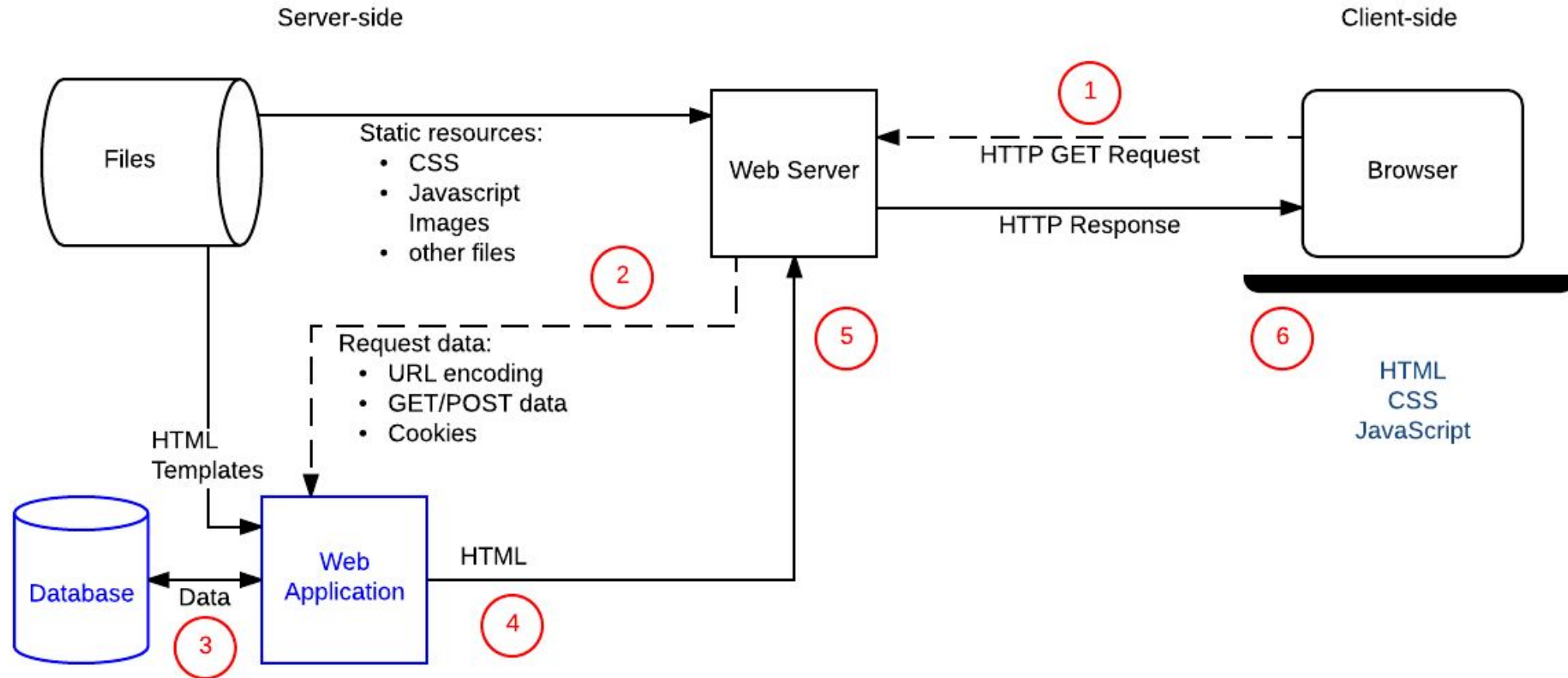
Static Server Architecture



Types of Servers - Functionality

1. **Static**
2. **Dynamic**
 - **Some Response Generated when Needed**
 - **Content Created from Backend Databases**
 - **Data Inserted into HTML Templates**

Dynamic Server Architecture



Types of Servers – Role Based



Server-Side Programming Languages

- **Python**
 - **Django**
 - **Flask**
- **Node JS**
- **Express JS**
- **Personal Home Page/ Hypertext Preprocessor – PHP**
- **Active Server Page Network Enabled Technologies – ASP .NET**

Introduction to Node JS

- **Open Source Server Run Time Environment**
- **Javascript**
 - **Allows Running JS on Server Side**
- **Asynchronous Programming**
- **Tasks of Server-Side**

Why Node JS?

- **PHP and ASP .net**
 - **Sends the task to the computer's file system.**
 - **Waits while the file system opens and reads the file.**
 - **Returns the content to the client.**
 - **Ready to handle the next request.**
- **Node JS**
 - **Sends the task to the computer's file system.**
 - **Ready to handle the next request.**
 - **The server returns the content to the client when the file open read/write is complete.**
- **Node JS – MERN Stack**

1. **Evan Hahn, “Express in Action: Writing, Building, and Testing Node.js Applications” – 15 April 2016, First Edition.**
2. **Brad Dayley, Brendan Dayley, Caleb Dayley, “Node.js, MongoDB and Angular Web Development: The definitive guide to using the MEAN stack to build web applications”, Pearson Education, Second Edition, 2017.**
3. **Greg Lim, “Beginning MERN Stack: Build and Deploy a Full Stack MongoDB, Express, React, Node.js App,” 2021.**
4. **Online Sources**
 - [Introduction to the server side](#)
 - [Node.js Tutorial \(w3schools.com\)](#)

THANK YOU

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— UNIVERSITY —
CHENNAI